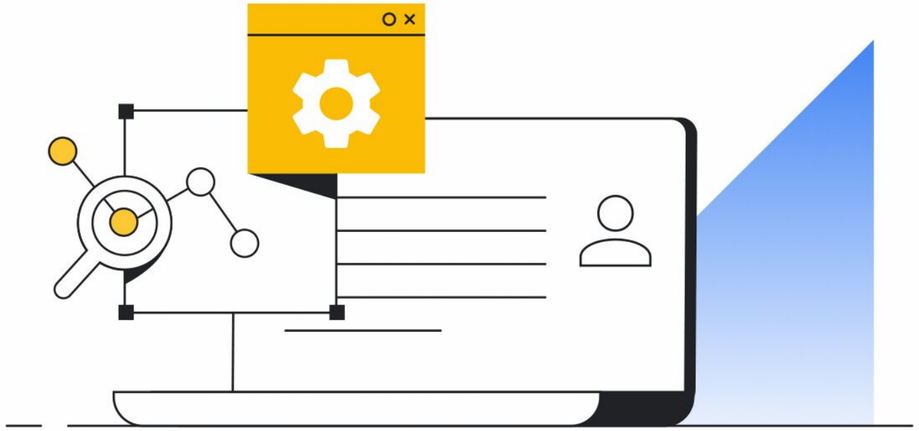
Google Cloud

Coexistence



Course

Google Workspace – Work Transformation

The information in this presentation is classified:

Google confidential & proprietary

⚠️ This presentation is shared with you under NDA.

- Do **not** record or take screenshots of this presentation.
- Do **not** share or otherwise distribute the information in this presentation with anyone **inside** or **outside** of your organization.

Thank you!

Proprietary + Confidential

Session logistics

Questions

In Google Meet, click the raise hand button or add your question to the Q&A section.

Answers may be deferred until the end of the session.

Slide availability

These slides are available in the Student Lecture section of your Qwiklabs classroom.

Recording

The session is **not** recorded.

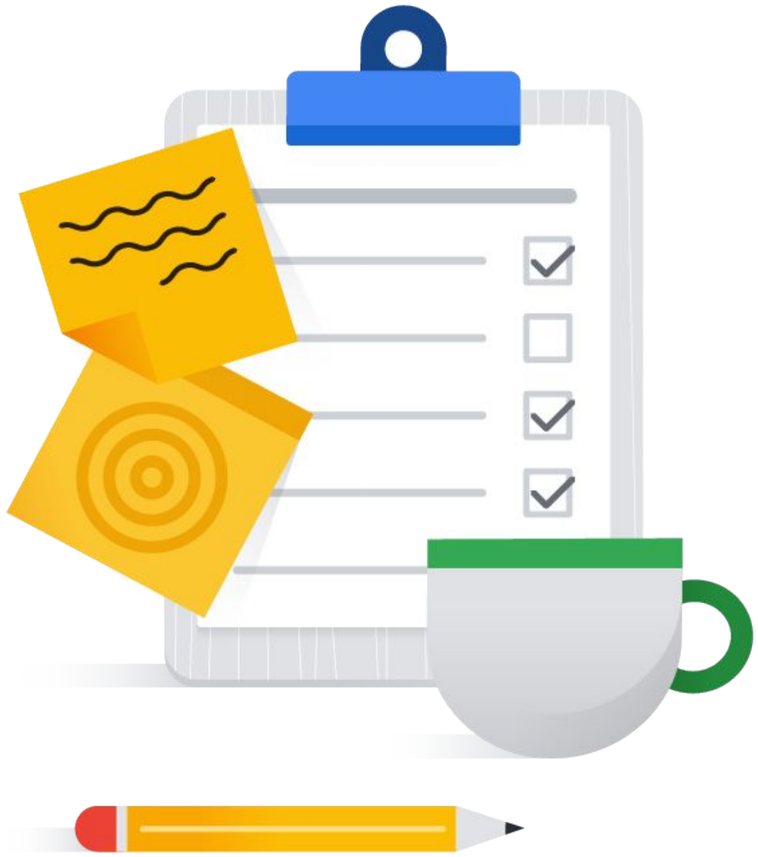
You are **not permitted** to record this session either.

Chat

As Google Meet does not have persistent chat, you will lose chat history if you get disconnected. Save URLs as they appear.

Module topics

01	Concepts
02	Strategy
03	Tools
04	Best practices and watchpoints



Outcomes

01 Understand the concept of ‘coexistence’

02 Describe the short-term and long-term aspects of coexistence

03 Explore the coexistence strategy

04 Use tools for managing coexistence

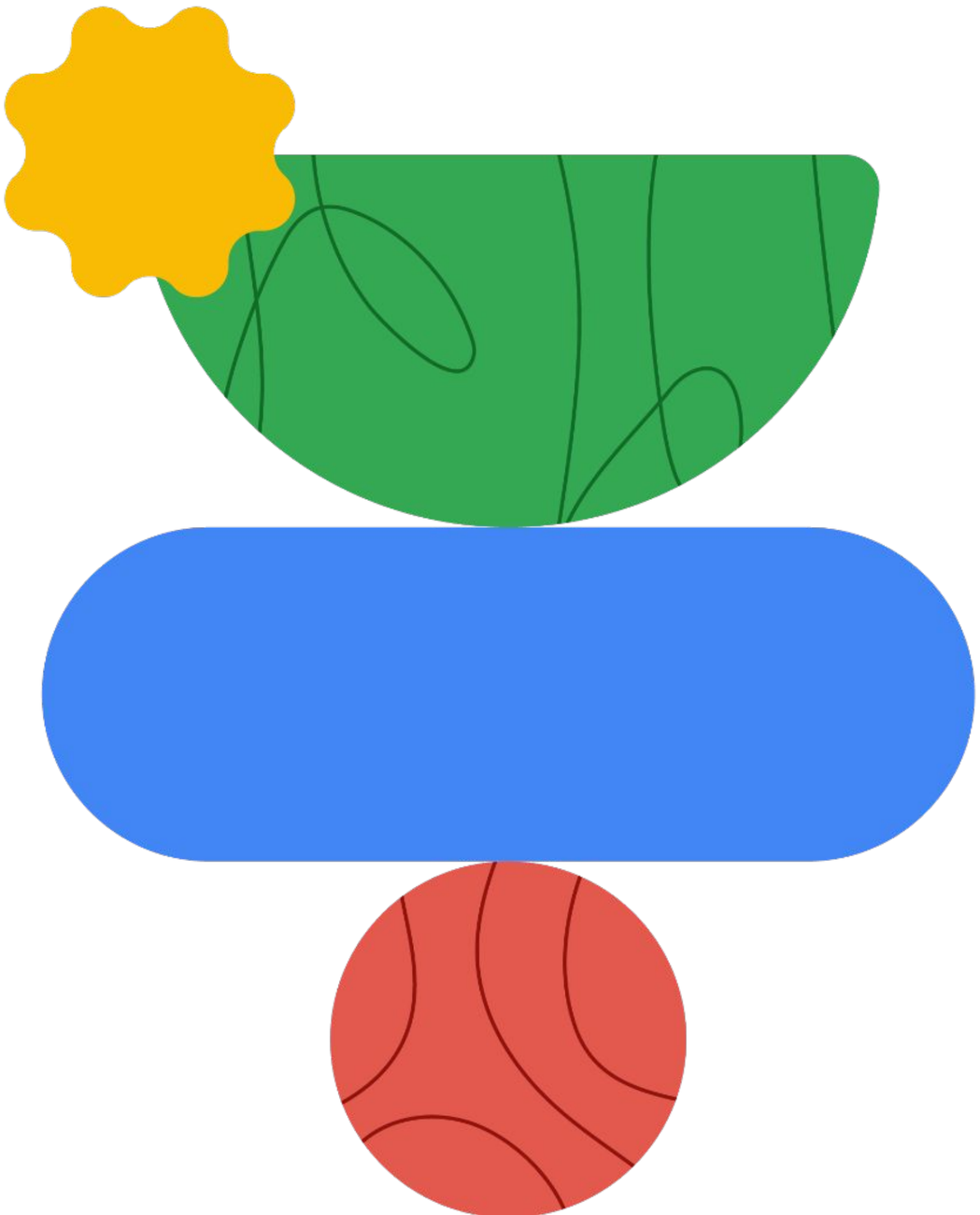
05 Explain how Google Workspace works during coexistence

Outcomes

06 Understand Groups provisioning during coexistence

07 Discuss best practices

Concepts



Proprietary + Confidential

Coexistence is when a newly-deployed system lives side-by-side with one or more legacy systems. It requires expertise and effort.

Proprietary + Confidential

Reasons for migrating

Robustness and security

Cloud-first organizations keep their IT infrastructure in the cloud for enhanced security.

Scalability

Cloud infrastructure scales on demand, providing a responsive and cost-effective IT capacity.

Collaboration and innovation

Cloud infrastructure enables seamless collaboration and resource sharing across organizations, irrespective of location.

Proprietary + Confidential

Types of coexistence

01

Short-term internal coexistence
Is necessary during deployment to ensure uninterrupted workflow as people gradually migrate from the legacy system to the new system.

02

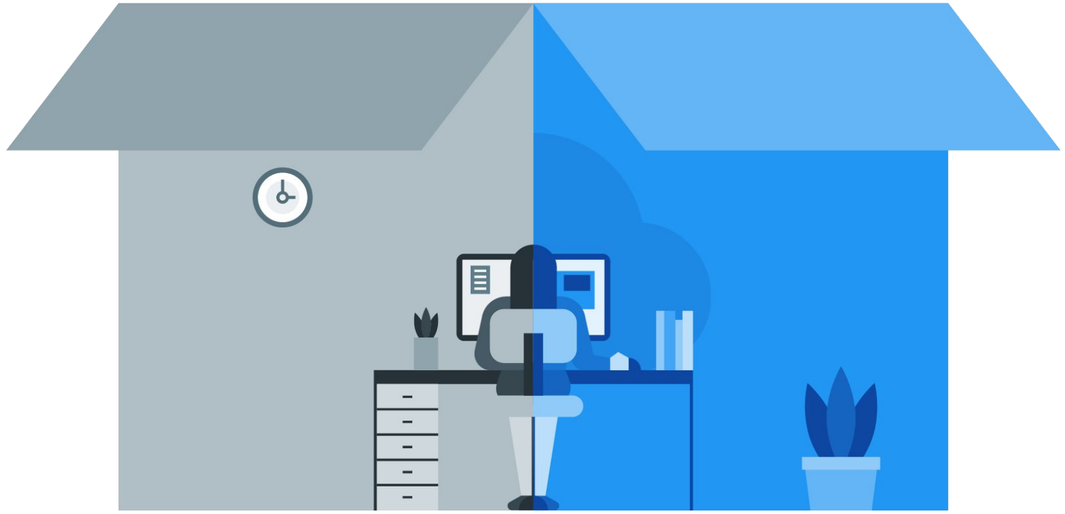
Long-term internal coexistence
Is not recommended, but may be required in certain circumstances such as maintaining a legacy application that will not work with Google Workspace.

03

Longer-term external coexistence
Is for situations where the organization works with external bodies that do not use Google Workspace.

Proprietary + Confidential

Short-term/long-term and internal/external coexistence



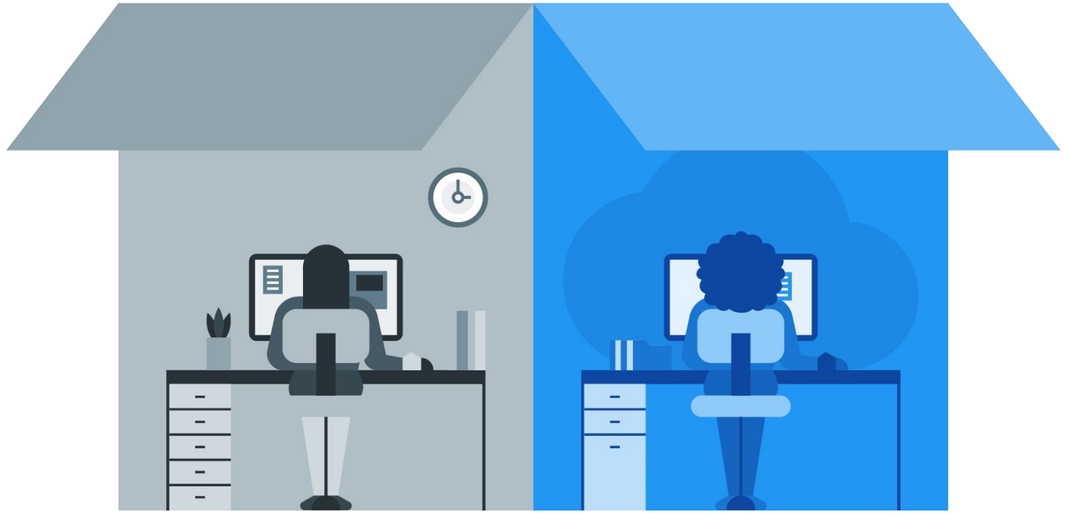
Short-term, internal coexistence.

Recommended

- Part of the transition plan
- Unavoidable during a 3-phase deployment

Proprietary + Confidential

Short-term/long-term and internal/external coexistence



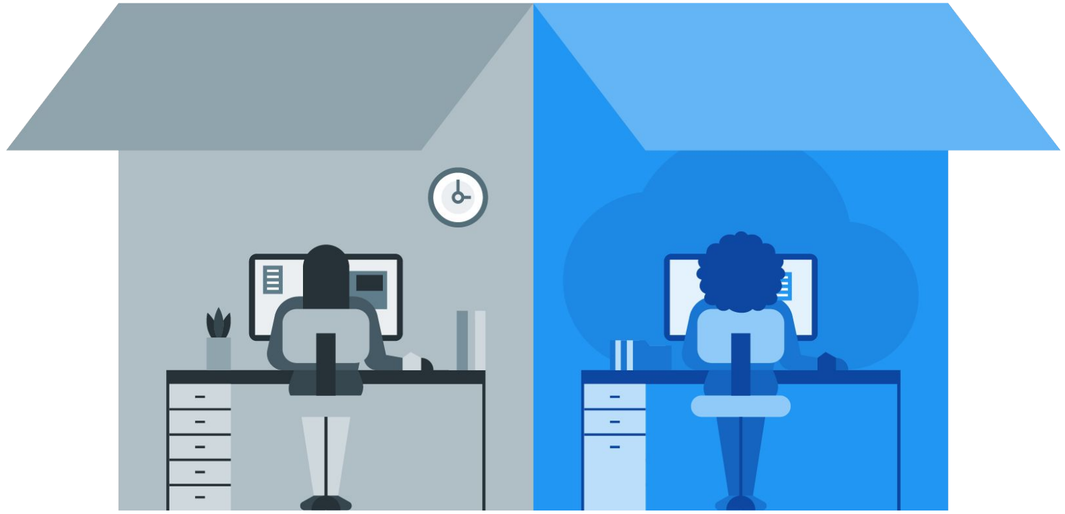
Long-term, internal coexistence.

Avoid

- Impact on admin and end user can jeopardize confidence
- Limits transformation opportunities
- Tools solve only some interoperability issues

Proprietary + Confidential

Short-term/long-term and internal/external coexistence



Long-term, external coexistence.

Not an issue

Long-term external coexistence is not an issue as Google Workspace is designed for interoperability with external applications and platforms.

Strategy

Proprietary + Confidential

Impact and mitigation

The longer the coexistence period, the greater the impact

Impact

- Users and resources on multiple systems
- Dedicated admin tasks and maintenance
- May require long-term infrastructure changes
- Limits transformation opportunities
- Perception issues and misappropriated frustrations

It is not possible to fully address all impacts of coexistence

Mitigation

- Limit the duration as much as possible
- Engage partner support and expertise
- Utilize Google first-party tools and applications
- Focus on change management

Proprietary + Confidential

Coexistence duration

Technical team

Duration: Entire deployment timeline

Responsible for:

- Configuring and maintaining both systems
- Addressing technical challenges
- Ensuring a smooth transition for end-users
- Fine-tune configurations before the full cutover

End-users

Duration: approx. 1 month

- This period begins after the Early Adopters (EA) Go Live, when EA users start actively using Google Workspace (GW).
- It ends with the Go Live for Global Go Live (GGL) users, when they are transitioned from the legacy system to GW.

Proprietary + Confidential

Coexistence duration

Coexistence for technical team

Phases / Workstreams	Excite (aka Sales)	Enable (aka Deployment) 3 or 4 months			Expand	Embed
		Core IT • 1 month • Max 25 technical users from the IT department	**Early Adopters** • 1 or 2 month(s) • 5% to 10% users representing different roles, departments and locations	**Global Go Live** • 1 month • Everyone else Coexistence for business users		
Governance						
People (Change Mgmt)						
Process (Change Mgmt)						
Technology						
Support						

Go Live for Core IT users

Go Live for EA users

Go Live for GGL users

Proprietary + Confidential

Coexistence should be used strategically

Deployment	Sustainability	Impact	Support tools
• Unavoidable • Define and guide • Minimize duration	• Short-term • Difficult to support • Costly	• Multiple systems in use • Duplicated admin tasks • Limited transformation opportunities	• Calendar • Exchange 2010+ • Office 365 • Meet - 3rd party offering

Coexistence

Proprietary + Confidential

Long-term: Mail

Drawbacks

Actions

Known issues

Decisions

Proprietary + Confidential

Long-term: Mail

Drawbacks

- Creates a black box for Google support
- Difficult to manage/support users in multiple environments
- User experience differences between desktop client and web browser

Proprietary + Confidential

Long-term: Mail

Drawbacks

Known issues

Decisions

Actions

- Determine what configuration changes are necessary to ensure mail flow is uninterrupted for all users
- Ensure on-premises systems properly authenticate email messages

Proprietary + Confidential

Long-term: Mail

Drawbacks

Actions

Decisions

Known issues

- Possible issues with calendar invites and group expansion
- Possible issues with multiple spam/compliance systems running in parallel

Proprietary + Confidential

Long-term: Mail

Drawbacks

Actions

Known issues

Decisions

- Where will mail route first for best performance?
- When will shared mailboxes be migrated?
- Will users be able to access mail via legacy system past go-live?

Proprietary + Confidential

Long-term: Mail

Drawbacks

Actions

Known issues

Decisions

Proprietary + Confidential

Long-term: Calendar

Drawbacks

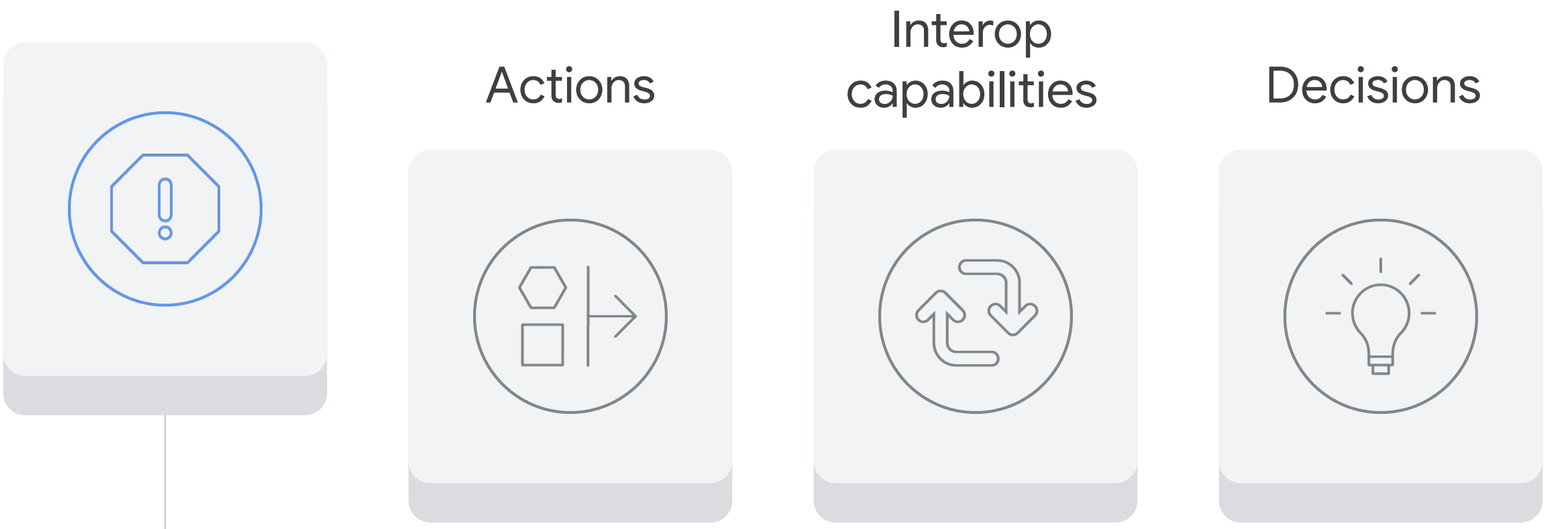
Actions

Interop capabilities

Decisions

Proprietary + Confidential

Long-term: Calendar



Drawbacks

- Google's Calendar Interop tool only supports some platforms and one external directory
- Not leveraging the interop tool may require users to manage multiple systems

Proprietary + Confidential

Long-term: Calendar

Drawbacks

Interop capabilities

Decisions

Actions

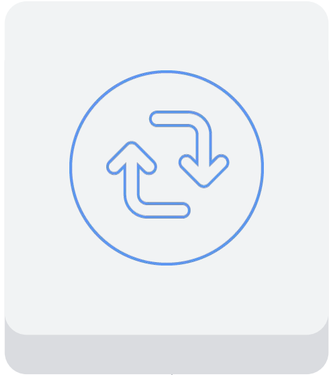
- Do not enable Google Calendar for users in another calendaring system
- Perform platform specific requirements gathering

Proprietary + Confidential

Long-term: Calendar

Drawbacks

Actions



Decisions

Interop capabilities

- Sync attendee status between platforms
- Booking calendar resources

Proprietary + Confidential

Long-term: Calendar

Drawbacks

Actions

Interop capabilities

Decisions

- Where will resources live and how do you ensure all users can book them?
- Is there a need to investigate interop with other video conferencing tools?

Proprietary + Confidential

Long-term: Calendar

Drawbacks

Actions

Interop capabilities

Decisions

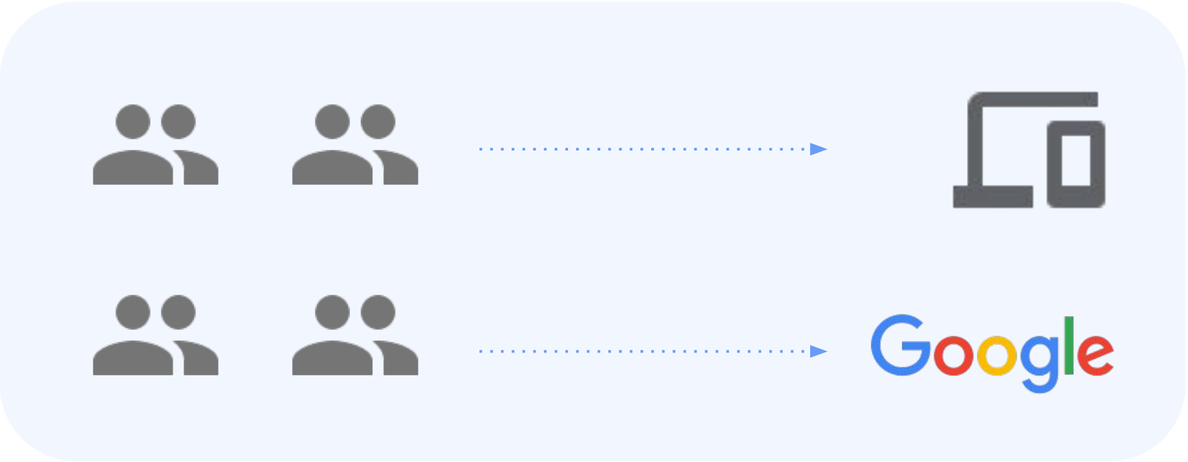
Proprietary + Confidential

Reserving calendar resources

Option 1

Migrate room resources that are only used by Google Workspace users.

E.g. building of department-specific.



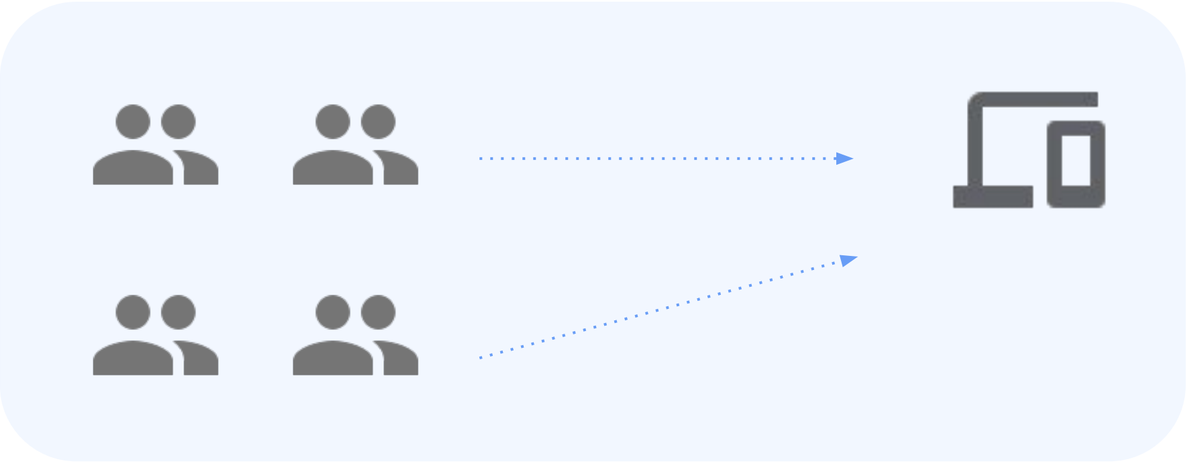
Proprietary + Confidential

Reserving calendar resources

Option 2

Instruct all users to reserve resources

in the legacy system until the global go-live.

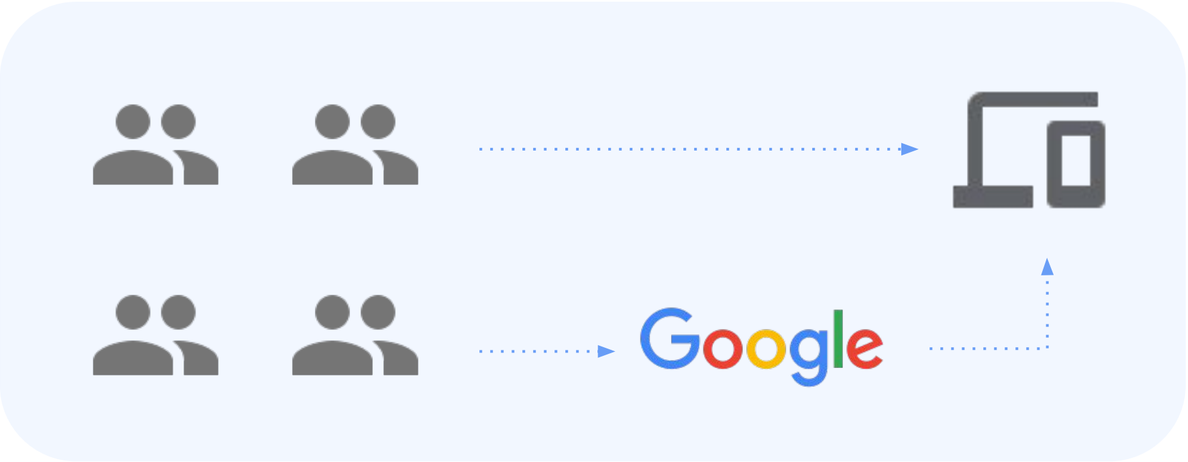


Proprietary + Confidential

Reserving calendar resources

Option 3

Calendar Interop tool allows users to leverage their respective environments, while rooms remain in Microsoft Exchange.



Strategy

Proprietary + Confidential

Long-term: Google Meet

Drawbacks

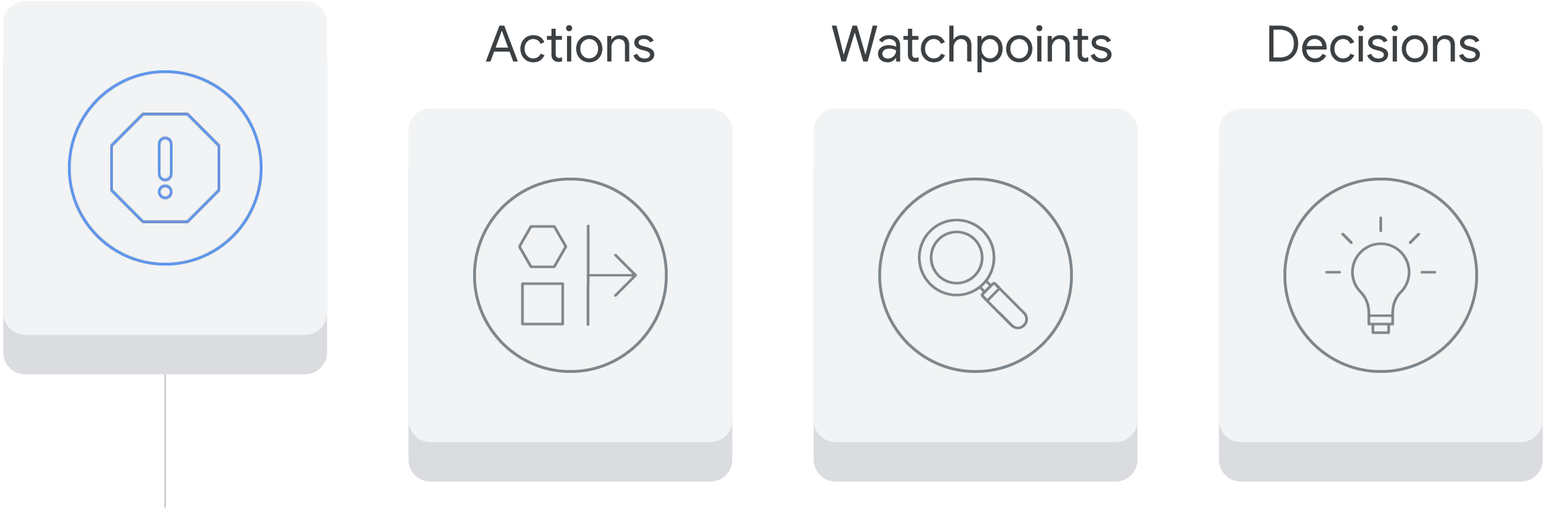
Actions

Watchpoints

Decisions

Proprietary + Confidential

Long-term: Google Meet



Drawbacks

- Requires additional infrastructure
- Licenses need to be purchased

Proprietary + Confidential

Long-term: Google Meet

Drawbacks

Watchpoints

Decisions

Actions

- Perform platform specific requirements gathering
- Review third-party interop solutions (for example, Pexip)

Proprietary + Confidential

Long-term: Google Meet

Drawbacks

Actions

Decisions

Watchpoints

- Allows for third-party clients to connect to Google Meet meetings, but not the other way around
- Split support process between Google and third-party vendor

Proprietary + Confidential

Long-term: Google Meet

Drawbacks

Actions

Watchpoints

Decisions

- Where and how to host and deploy third-party software?

Proprietary + Confidential

Long-term: Office

Drawbacks

Actions

Known issues

Decisions

Proprietary + Confidential

Long-term: Office

Actions

Known issues

Decisions

Drawbacks

- May require Drive for desktop to enable interop with Office desktop clients
- Maintaining Office licenses could be costly
- Users of complex macros may not have a migration path

Proprietary + Confidential

Long-term: Office

Drawbacks

Known issues

Decisions

Actions

- Determine which users require Office applications
- Google recommends identifying true usage of applications
  - Applications exist for measuring utilization

Proprietary + Confidential

Long-term: Office

Drawbacks

Actions

Decisions

Known issues

- Document conversion supports most, but not all, functionality and formatting
- Apps Script cannot fully replicate all VBA functions

Proprietary + Confidential

Long-term: Office

Drawbacks

Actions

Known issues

Decisions

- Do all users need to retain access to Office or can it be a smaller subset?
- Where and how to deploy Drive for Desktop?

Proprietary + Confidential

Google Groups

Proprietary + Confidential

Provisioning Groups

Delay Group provisioning

- Continue using legacy groups during initial deployment.
- Delays addressing group synchronization challenges.

Early Group provisioning

- Create Google Groups early in the deployment
- Tackle synchronization issues in real-time
- Prevents issues from escalating into larger problems later at a later stage

Proprietary + Confidential

Core IT

01

Group inventory and mapping

Start creating an inventory and map of your existing legacy groups.

02

Promote collaboration tools

Help your project team to start using the collaboration tools in Google Workspace.

03

Initial synchronization

Install Google Cloud Directory Sync and complete an initial synchronization simulation.

04

Provisioning Groups

Focus on thorough planning and testing during the Core IT phase.

Proprietary + Confidential

Early adopters (EA)

Synchronize groups and their membership to Google Workspace from the legacy system.

Users can use Google Groups to share content during this phase.

Proprietary + Confidential

Group expansion

Key watchpoint

The potential for increased email volume when groups are invited to calendar events, due to Group expansion.

Legacy platform user invites the group → The legacy system expands the group

Google Workspace user invites the group → Expansion occurs within Google Workspace

Proprietary + Confidential

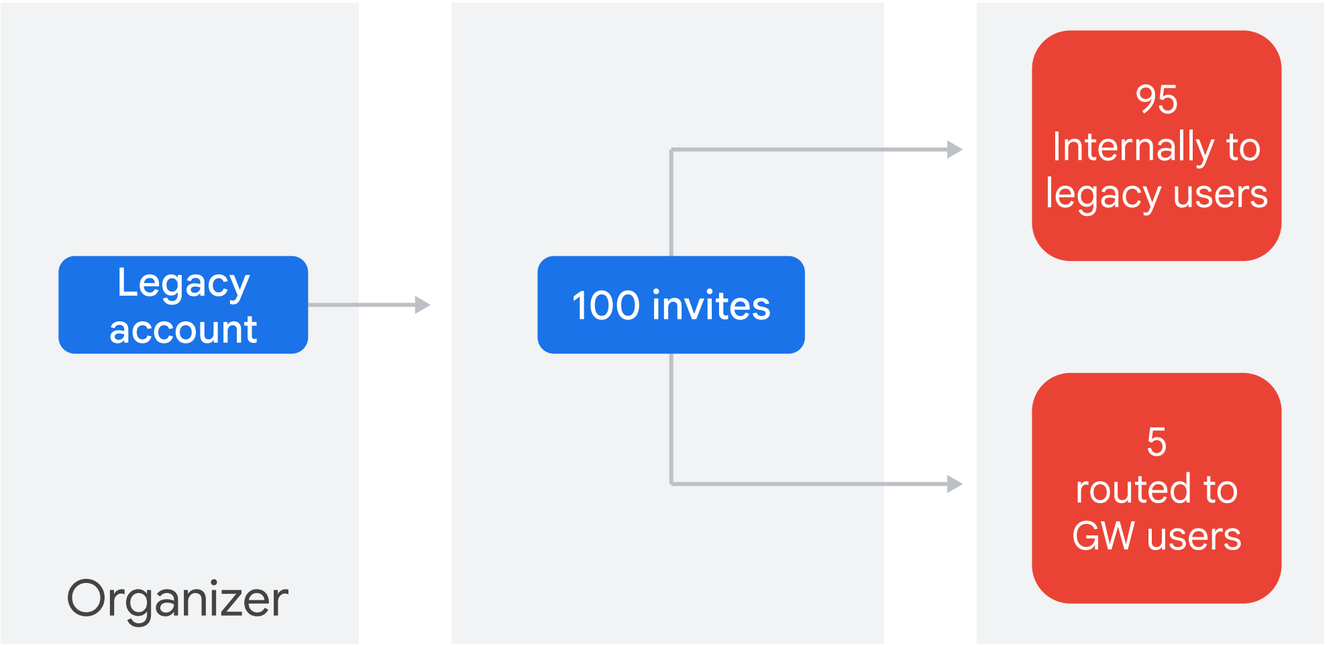
Group expansion: Legacy account is the organizer

marketing@yourcustomer.com

100 members

5 early adopters

95 Global go-live



Expansion occurs within Legacy platform

Proprietary + Confidential

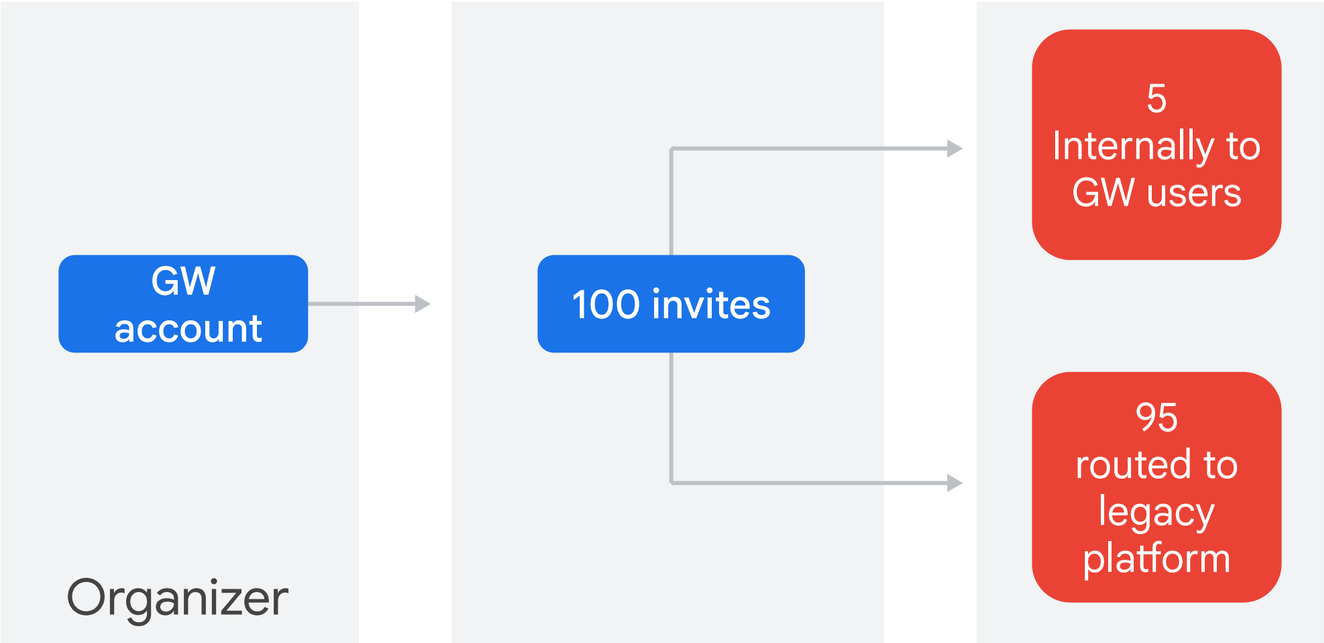
Group expansion: GW account is the organizer

marketing@yourcustomer.com

100 members

5 early adopters

95 Global go-live



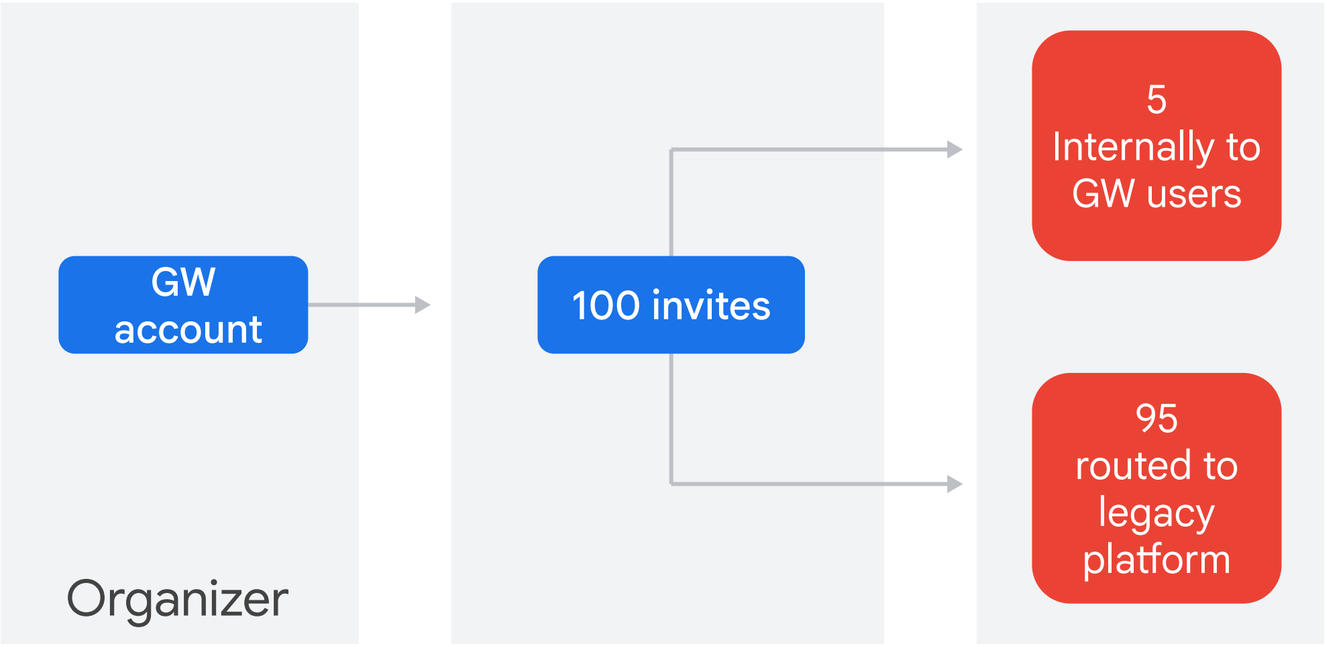
Expansion occurs within Google Workspace

Proprietary + Confidential

Group expansion: GW account is the organizer

Key watchpoint

Prevent email delivery to partially populated groups by synchronizing groups to Google Workspace before you point the MX records to Google.



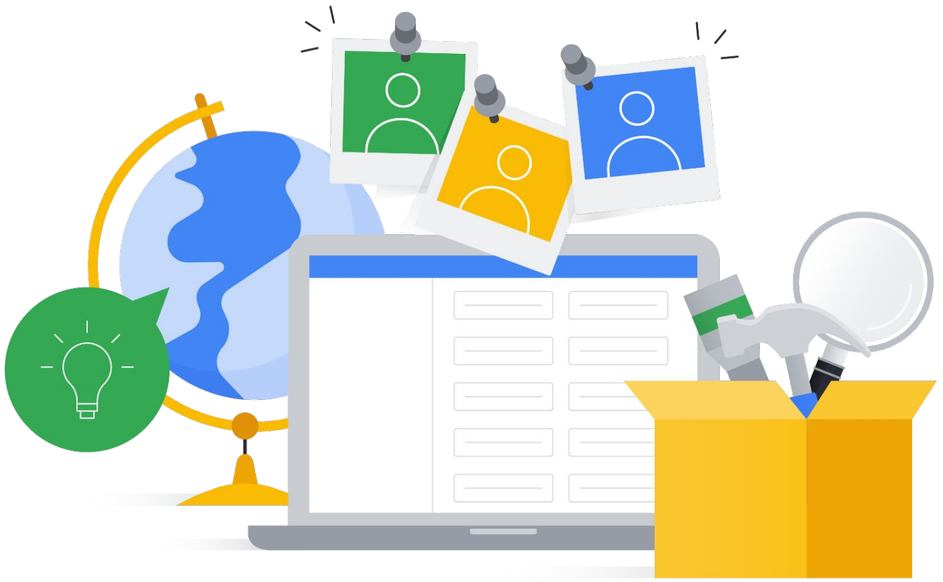
Expansion occurs within Google Workspace

Proprietary + Confidential

Global go-live (GGL)

All legacy groups provisioned and synced, most migration challenges met.

- Migrate public folders to Google Groups
- Convert shared mailboxes to Google Groups



Tools

Proprietary + Confidential

Be strategic

Don't fall into the trap of using tools just because you have them.

The goal is to select the most effective solutions for your specific needs, not to overcomplicate the process.

Proprietary + Confidential

Coexistence and interoperability

Mail

- Requires decisions about mail routing
- Can cause issues with mail security and compliance systems

Calendar

- Only supported on some platforms
- May need to maintain DMZ servers
- Challenges booking resources

Meet

- Requires additional infrastructure
- Third party can connect to Meet sessions, but not the other way around

Proprietary + Confidential

Calendar interop for Microsoft Exchange

Cloud-based interop service for calendar services between Google and Microsoft Exchange.

Supported versions	Microsoft Exchange 2013 or 2016 or Exchange Online (Microsoft 365)
User management	For users with accounts in both systems: • **Exchange calendar users**: Google Calendar service must be disabled • **Google calendar users**: Hide Exchange mailboxes, and set forwarding contacts to be visible in the global address list
Watchpoint	Supports only one external source of availability information at a time
Resource management	Interop allows Google users to book room resources in Microsoft Exchange
Support	Developed and supported by Google

Proprietary + Confidential

Google Drive for desktop

Stream My Drive and Shared Drive files directly from the cloud.

Supported versions	• Windows (64-bit only):
	  ○ Windows 10 and up
	  ○ Windows Server 2016 and up
	• Mac:
	  ○ macOS Catalina 10.15.7 and up

Sync capabilities	Stream Drive files on demand, or make them available for offline access (Mirror)
Coexistence	Allows users to see when someone is editing a shared Microsoft Office file
Watchpoint	Requires certain hosts be allowed through network firewalls
Support	Developed and supported by Google

Proprietary + Confidential

Google Meet add-in for Microsoft Outlook

Add a Google Meet video call directly into your Outlook event or email.

Supported versions	Outlook 2016 or 2019 for Windows, Outlook 2016 for Mac, or Outlook on the web.
Capability	Allows Microsoft Exchange users to book Google Meet conferences from Outlook
Watchpoint	Not supported on mobile devices
Support	Bundled with Drive for Desktop client

Proprietary + Confidential

Meet interoperability with Pexip

Web user from computer

Google Meet for Android

Chromebox for meetings

Pexip Infinity Distributed Gateway provides any-to-any video interoperability with Google Meet.

Google Meet

]pexip[

H.323/SIP video endpoints and Skype for Business dialling to Google Meet via Virtual Reception and Pexip Fusion gateway.

Proprietary + Confidential

Pexip infinity platform for Google Meet

Enables native interoperability between Hangouts Meet and SIP/H.323 endpoints

Supported endpoints	SIP/H.323 endpoints (Cisco, Polycom, Lifesize, and more) and Skype for Business
Licenses	Licenses need to be purchased from Pexip or Pexip partner
Watchpoint	Allows for third-party clients to connect to Google Meet meetings, but not the other way around
Watchpoint	Requires the hosting and deployment of third-party software
Watchpoint	Split support process between Google and Pexip/partner

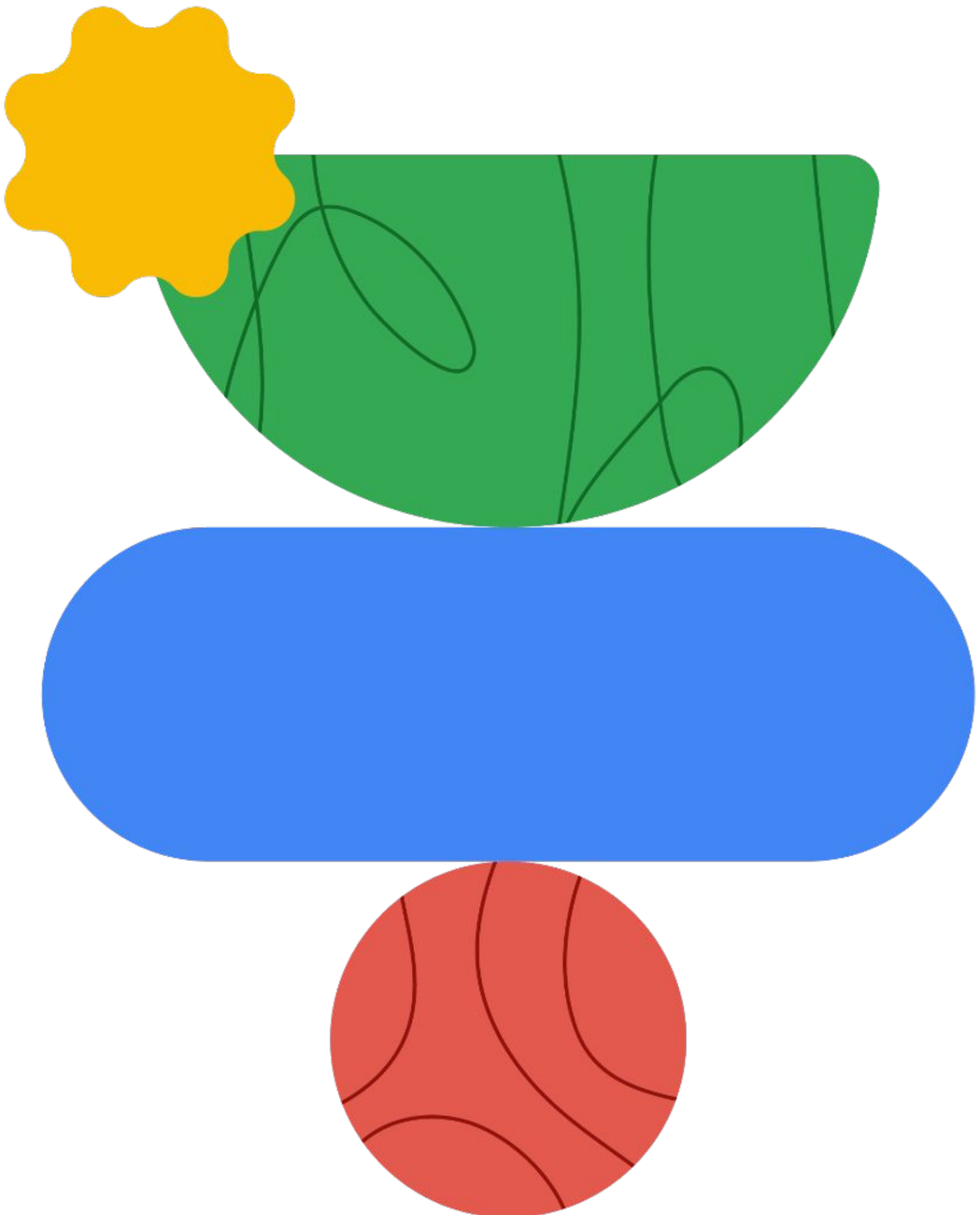
Proprietary + Confidential

Google Workspace Sync for Microsoft Outlook

Synchronize mail, calendar, contacts, notes, and tasks between Google Workspace and Outlook.

System Requirements	Microsoft Windows - 32- and 64-bit versions of Windows Vista*, 7*, 8*, 10, and 11 Microsoft Outlook • Outlook 2003 SP3* or 2007 SP2* (or SP1 with hotfix)* • 32- and 64-bit versions of Outlook 2010*, 2013*, 2016, 2019, and 2021 • Outlook from Microsoft 365 desktop app (not web version)
Sync Details	Allows Gmail users to use Outlook for email, calendar, contacts, notes, and tasks
Watchpoint	Native Google Workspace features **are not available** in Outlook (for example, smart compose) Some Outlook features are not supported (for example, rich text format emails) Sync client introduces complexity and potential failure points
Recommendation	Use only as a stopgap solution where thick client integrations are required Guide users to adopt Gmail web UI and avoid long term reliance on Outlook
Support	Developed and supported by Google

Best practices and watchpoints



Proprietary + Confidential

Coexistence dos and don’ts

Avoid

- Multiple mailboxes
- Multiple calendars
- Individual user choice for email system
- Long-term email coexistence
- GWSMO (Google Workspace Sync for Microsoft Outlook)

Proprietary + Confidential

Coexistence dos and don’ts

Avoid

- Multiple mailboxes
- Multiple calendars
- Individual user choice for email system
- Long-term email coexistence
- GWSMO (Google Workspace Sync for Microsoft Outlook)

Proceed with caution

- Deployment coexistence
- Split delivery email routing
- Google Drive for desktop client helps Microsoft users adopt Google services
- Calendar interop service facilitates availability sharing and resource booking

Proprietary + Confidential

Coexistence dos and don’ts

Avoid

- Multiple mailboxes
- Multiple calendars
- Individual user choice for email system
- Long-term email coexistence
- GWSMO (Google Workspace Sync for Microsoft Outlook)

Proceed with caution

- Deployment coexistence
- Split delivery email routing
- Google Drive for desktop client helps Microsoft users adopt Google services
- Calendar interop service facilitates availability sharing and resource booking

Encourage

- Collaboration during deployment
- Google Meet availability
- Visitor sharing to non-Google Workspace users

Proprietary + Confidential

Change management and recommendations

It's critical to communicate clear instructions to coexistence users throughout deployment.

- When choosing core IT and early adopters, ensure users understand possible need for multiple systems
- Explain that coexistence is temporary and part of the transition plan
- Tailor training to discuss workarounds for coexistence watchpoints and differences in environments
- Ensure support desk is well-trained to manage queries
- Administrative assistants should be migrated with their executives to ease coexistence challenges

Proprietary + Confidential

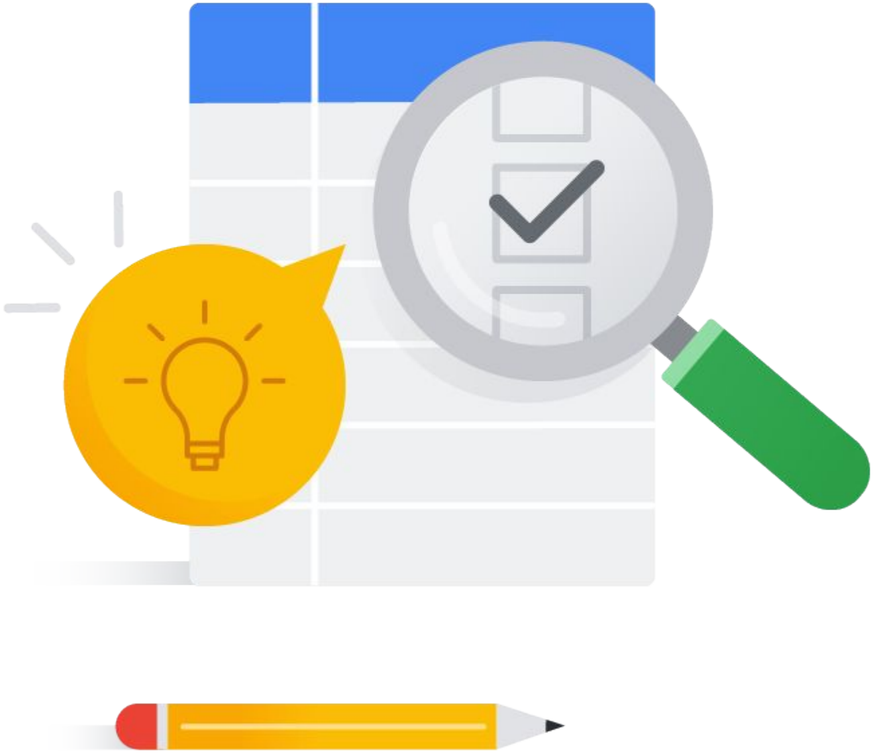
Lab: Work Transformation Part 4: Exploring Google Docs & Microsoft Office Coexistence

30:00

20 min

Learning objectives

- Install and configure Google Drive for desktop
- Explore how it coexists alongside Microsoft Office.



Proprietary + Confidential

References

https://support.google.com/

Google Cloud

About GWSMO

https://support.google.com/

Google Cloud

Use Google Drive for Desktop

https://apps.google.com/

Google Cloud

Add Meet video meetings to Outlook

https://support.google.com/

Google Cloud

Share content with a group

Google Cloud